FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2022

CONCEPT RECAPITULATION TEST – II

PAPER –1

TEST DATE: 30-06-2022

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 57 questions.
- Each subject (PCM) has 19 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.

Section – A ***(01 – 04, 20 – 23, 39 – 42)*****:** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

Section – A ***(05 –10, 24 – 29, 43 – 48)*****:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – B ***(11 – 13, 30 – 32, 49 – 51)*****:** This section contains **NINE (09)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.**

Section – C ***(14 – 19, 33 – 38, 52 – 57)*****:** This section contains **NINE (09)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+2	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

Physics

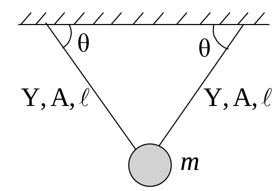
PART – I

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

1. During walking, work done by fraction is:
   (A) Positive
   (B) Negative
   (C) Zero
   (D) None of these

2. The time period of small vertical oscillations of pendulum is: (A pendulum bob is hanging in equilibrium with the help of two identical elastic wires as shown in the figure.)



   (A) $2\pi\sqrt{m\ell / YA}$
   (B) $2\pi\sqrt{m\ell / 2YA}$
   (C) $\pi\sqrt{m\ell / YA}$
   (D) None of these

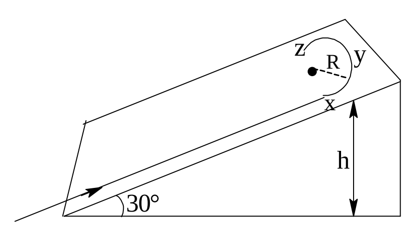
3. The average degree of freedom per molecule of a gas is 6. The gas performs 25J work, while expanding at constant pressure. The heat absorbed by the gas is:
   (A) 75J
   (B) 100J
   (C) 150J
   (D) 125J

4. The radioactive sources A and B have half lives of 2hr and 4hr respectively, initially contain the same number of radioactive atoms. At the end of 2 hours, their rates of distintegration are in the ratio:
   (A) 4:1
   (B) 2:1
   (C) $\sqrt{2}:1$
   (D) 1:1

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A student skates up a ramp that makes an angle 30° with the horizontal. He/she starts (as shown in the figure) at the bottom of the ramp with speed v_0 and wants to turn around over a semicircular path xyz of radius R during which he/she reaches a maximum height h (at point y) from the ground as shown in the figure. Assume that the energy loss is negligible and the force required for this turn at the highest point is provided by his/her weight only. Then (g is the acceleration due to gravity)



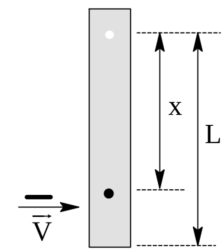
   (A) $v_0^2 - 2gh = \frac{1}{2}gR$

(B) $v_0^2 - 2gh = \frac{\sqrt{3}}{2}gR$

(C) The centripetal force required at points x and z is zero

(D) The centripetal force required is maximum at points x and z

6. A rod of mass m and length L, pivoted at one of its ends, is hanging vertically. A bullet of the same mass moving at speed v strikes the rod horizontally at a distance x from its pivoted end and gets embedded in it. The combined system now rotates with angular speed ω about the pivot. The maximum angular speed ω_M is achieved for $x = x_M$. Then:



(A) $\omega = \frac{3vx}{L^2 + 3x^2}$

(B) $\omega = \frac{12vx}{L^2 + 12x^2}$

(C) $x_M = \frac{L}{\sqrt{3}}$

(D) $\omega_M = \frac{V}{2L}\sqrt{3}$

7. In an X-ray tube, electrons emitted from a filament (cathode) carrying current I hit a target (anode) at a distance d from the cathode. The target is kept at a potential V higher than the cathode resulting in emission of continuous and characteristic X-rays. If the filament current I is decreased to $I/2$, the potential difference V is increased to 2V, and the separation distance d is reduced to $d/2$, then:

(A) the cut-off wavelength will reduce to half, and the wavelengths of the characteristic X-rays will remain the same

(B) the cut-off wavelength as well as the wavelengths of the characteristic X-rays will remain the same

(C) the cut-off wavelength will reduce to half, and the intensities of all the X-rays will decrease

(D) the cut-off wavelength will become two times larger, and the intensity of all the X-rays will decrease

8. Two identical non-conducting solid spheres of same mass and charge are suspended in air from a common point by two non-conducting, massless strings of same length. At equilibrium, the angle between the strings is α. The spheres are now immersed in a dielectric liquid of density 800 kg m^{-3} and dielectric constant 21. If the angle between the strings remains the same after the immersion, then

(A) electric force between the spheres remains unchanged

(B) electric force between the spheres reduces

(C) mass density of the spheres is 840 kg m^{-3}

(D) the tension in the strings holding the spheres remains unchanged

9. Starting at time t = 0 from the origin with speed 1 ms^{-1}, a particle follows a two-dimensional trajectory in the x-y plane so that its coordinates are related by the equation $y = \frac{x^2}{2}$. The x and y components of its acceleration are denoted by a_x and a_y, respectively. Then:

(A) $a_x = 1\,ms^{-2}$ implies that when the particle is at the origin, $a_y = 1\,ms^{-2}$

(B) $a_x = 0$ implies $a_y = 1\,ms^{-2}$ at all times.
(C) at t = 0, the particle's velocity points in the x-direction
(D) $a_x = 0$ implies that at t = 1s, the angle between the particle's velocity and the x axis is 45°

10. A source is moving in a circle given by $x^2+y^2=R^2$ with a constant speed $\frac{330\pi}{6}$ m/s in clockwise sense. A detector is at rest at (2R, 0). Find the coordinates of the source at the instant when the detector records the same frequency as that of the source (speed of sound = 330 m/s):

(A) $\left(\frac{\sqrt{3}R}{2}, -\frac{R}{2}\right)$
(B) $\left(\frac{\sqrt{3}R}{2}, \frac{R}{2}\right)$
(C) $(0, R)$
(D) $(R, 0)$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

11. Consider one mole of helium gas enclosed in a container at initial pressure P_1 and volume V_1. It expands isothermally to volume $4V_1$. After this, the gas expands adiabatically and its volume becomes $32V_1$. The work done by the gas during isothermal and adiabatic expansion processes are W_{iso} and W_{adia}, respectively. If the ratio $\frac{W_{iso}}{W_{adia}} = \frac{16}{k} \ln 2$, then find the value of k?

12. A stationary tuning fork is in resonance with an air column in a pipe. If the tuning fork is moved with a speed of $2\,ms^{-1}$ in front of the open end of the pipe and parallel to it, the length of the pipe should be changed for the resonance to occur with the moving tuning fork. If the speed of sound in air is $320\,ms^{-1}$, the smallest value of the percentage change required in the length of the pipe is $\frac{k}{8}$, find the value of k?

13. A circular disc of radius R carries surface charge density $\sigma(r) = \sigma_0\left(1 - \frac{r}{R}\right)$, where σ_0 is a constant and r is the distance from the center of the disc. Electric flux through a large spherical surface that encloses the charged disc completely is ϕ_0. Electric flux through another spherical surface of radius $R/4$ and concentric with the disc is ϕ. If the ratio of $\frac{\phi_0}{\phi}$ is $\frac{32}{k}$, then find the value of k?

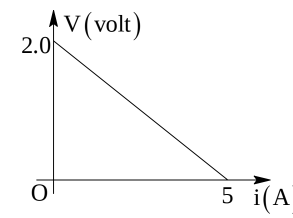
Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 14 and 15

Question Stem

The potential difference (V) between the positive and negative terminals of a cell changes with the current through it as shown in the graph



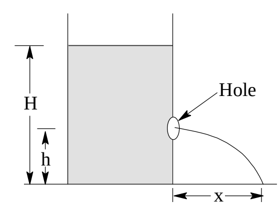
14. Find the internal resistance of the cell. (in ohm)

15. Find the emf of the cell (in volts)

Question Stem for Question Nos. 16 and 17

Question Stem

A tank placed on ground has water filled in it, up to a height H. A small hole is punched in its side wall at a height h from the bottom. Water jet strikes the ground at a horizontal distance x.
(Take H = 2 meter)



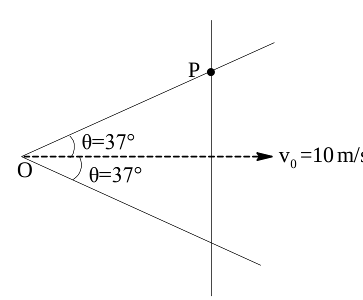
16. For what value (in meter) of h is x maximum?

17. For two value of h, we get same range (X). If one value of h is $H/4$, what is its other value? (in meter)

Question Stem for Question Nos. 18 and 19

Question Stem

A rod is translating on a V-shaped rail as shown in the figure at an instant.



18. Find the velocity (in m/s) of point of intersection P.

19. Find the angular velocity of intersection point P with respect to observer O.

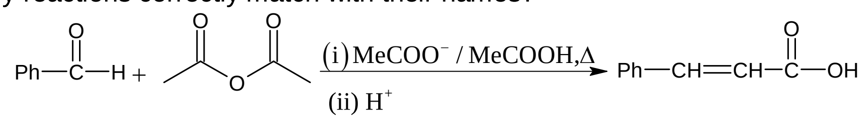
Chemistry

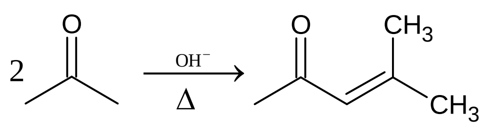
PART – II

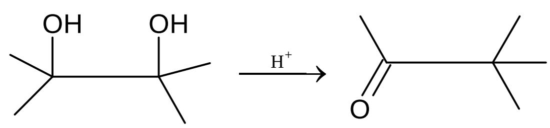
Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

20. Identify reactions correctly match with their names?

(A)  Perkin reaction

(B)  Cannizaro reaction

(C)  Friedel Crafts reaction

(D) $2\,H-\overset{\overset{\displaystyle O}{\|}}{C}-H \xrightarrow[\text{NaOH}]{\text{conc.}} \text{H-COO}^- + CH_3OH$ Haloform reaction

21. For the gas phase reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$ the equilibrium constant is $10^{-5}\,atm^{-2}$ at 300K. ΔG° for the reaction will be:
(A) $-28.72\,kJ$
(B) $+28.72\,kJ$
(C) $-14.3\,kJ$
(D) Zero

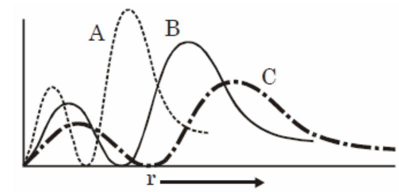
22. In salt analysis, cations of group IIIrd can be tested by adding certain reagents. Which is the correct combination for the above mentioned test?
(A) $Bi^{+3}, NH_4Cl / NH_4OH$
(B) $Fe^{+3}, NH_4Cl / NH_4OH$
(C) $Cr^{+3}, H_2S / HCl$
(D) $Al^{+3}, NH_4OH / H_2S$

23. In a titration, H_2O_2 is oxidised to O_2 by MnO_4^- in acidic medium. If 24 ml H_2O_2 requires 16ml of 0.1M MnO_4^-, then the volume strength of H_2O_2 will be:
(A) 0.167
(B) 1.89
(C) 3.72
(D) 0.93

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

24. Out of the following, which is not correct match for radial probability of finding the electron of 2s orbital?



(A) $A-H,\ B-He^{+},\ C-Li^{2+}$

(B) $A-He^{+},\ B-H,\ C-Li^{2+}$

(C) $A-Li^{2+},\ B-He^{+},\ C-H$

(D) $A-Li^{2+},\ B-H,\ C-He^{+}$

25. The compound Na_2IrCl_6 reacts with triphenylphosphine in diethylieneglycol in an atmosphere of CO to give $\left[IrCl(CO)(PPh_3)_2\right]$, known as 'Vaska's compound'.
(Atomic number of Ir = 77)
Which of the following statements is correct?
(A) The IUPAC name of the complex is carbonylchloridebis (triphenylphosphine) iridium(III)
(B) The hybridisation of the metal ion is sp^3
(C) The magnetic moment (spin only) of the complex is zero
(D) The complex shows geometrical as well as ionization isomerism

26. Choose the correct statement from the following:
(A) The slag obtained during extraction of iron is heavy and has lower melting point than that of metal
(B) At temperature below 983 K(approx) CO is chief reducing agent in blast furnance.
(C) In zone refining impurities moves in the direction of heater.
(D) Electrolytic reduction of Al_2O_3 is known as Hall-Heroult process.

27. Select the correct statement.
(A) Three moles of phenylhydrazine and one mole of aldose required to produce phenylosazone.
(B) Glucose form gluconic acid with Br_2/HOH
(C) Glucose form Glucaric acid with HNO_3
(D) Lowering in carbon chain can be done by Kiliani Fischer method.

28. Select the CORRECT statement.
(A) At Boyle's temperature a real gas behaves like an ideal gas irrespective of pressure.
(B) $\dfrac{T_{Boyle}}{T_{Critical}}=\dfrac{27}{8}$
(C) On increasing the temperature four times, collision frequency (Z_1) becomes double at constant volume.
(D) At high pressure Van der Waals constant 'b' dominates over 'a'.

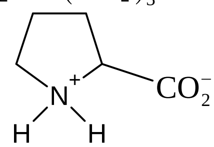
29. $HO_2C-(CH_2)_3-CH_2CO_2H \xrightarrow[2.\ NH_3(1eq.)]{1.1eq.\ of\ SOCl_2} P_1 \xrightarrow[+KOH]{Br_2} P_2 \xrightarrow[/red\,P]{Br_2} P_3 \xrightarrow{\Delta,\text{intramolecular SN2}} P_4$

Which are correct?

(A) P_1 is $H_2N-CO(-CH_2)_3-CH_2CO_2H$

(B) P_2 is $H_2N-(CH_2)_3-CH_2CO_2H$

(C) P_3 is $H_2N-(CH_2)_3-CH(Br)-CO_2H$

(D) P_4 is 

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

30. Xe react directly with F_2 in different condition (by volume mixture) of Xe & F_2 at suitable condition

Example: - $Xe + F_2 \longrightarrow XeF_2$

(2 : 1)

Ration of Xe and F_2 is given to make different Xenon $(\text{Fluoride})_n$ products.

$XeF_2 \Rightarrow Xe{:}F_2 = x : a = 1 : 0.5$

$XeF_4 \Rightarrow Xe{:}F_2 = x : b$

$XeF_6 \Rightarrow Xe{:}F_2 = x : c$

Find the value of $a+b+c$ when $x = 2$

31. Borax and kernite both are consist of $Na_2B_4O_7xH_2O$. Find sum of H_2O molecules in formula of borax & kernite.

32. When 0.2 g of acetic acid is added to 20 g of benzene, its freezing point decreases by 0.8°C. Find the % dimerisation of acetic acid in benzene, K_f (Benzene) $= 5$ K-kg mol.

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 33 and 34

Question Stem

A solid compound 'G', of formula $C_{15}H_{15}ON$, was insoluble in water, dilute HCl, of dilute NaOH. After prolonged heating of 'G' with aqueous NaOH, a liquid, 'H', was observed floating on the surface of the alkaline mixture. 'H' did not solidify upon cooling to room temperature; it was steam-distilled and separated. Acidification of the alkaline mixture with hydrochloric acid caused precipitation of a white solid, 'I'. 'I' is a paramethyl benzene derivative.
Compound 'H' was soluble in dilute HCl, and reacted with benzenesulfonyl chloride and excess KOH to give a base-insoluble solid, 'J'.
Compound 'I', m.p. 180°C, was soluble in aqueous $NaHCO_3$, and contained no nitrogen.
What were compound G, H, I and J?

33. The number of Carbon atoms in 'I' will be

34. The $\frac{\text{Molecular weight of 'H'}}{53.5}$ will be

Question Stem for Question Nos. 35 and 36

Question Stem

In coordination chemistry there are a variety of methods applied to find out the structure of complexes. One method involves treating the complex with known reagents and from the nature of reaction, the formula of the complex can be predicated. An isomer of the complex $Co(en)_2(H_2O)Cl_2Br$, on reaction with concentrated H_2SO_4 (dehydrating agent) it suffers loss in weight and on reaction with $AgNO_3$ solution it gives a white precipitate which is soluble in NH_3(aq.) [Atomic weight of Co = 59, Br = 80, Cl = 36]

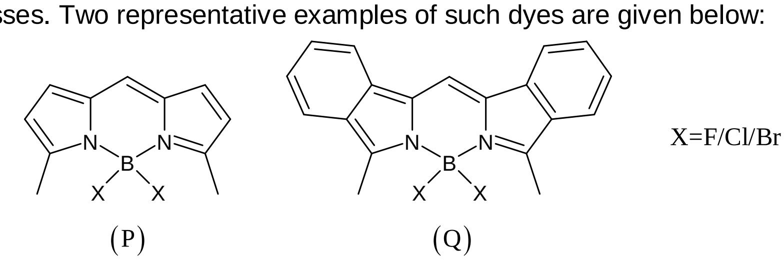
35. Number of geometrical isomers of the formula of the above original complex are (including the complex)

36. Molecular weight of cation of the above complex.

Question Stem for Question Nos. 37 and 38

Question Stem

Boron compounds are not only used for therapeutic but also for diagnostic applications. A class of pyrrole-containing boron compounds are used as dyes to stain and image various (transparent) organelles inside the living cells, which are ordinarily difficult to see. The $B-N$ bonds in these compounds are very stable even under biochemical condition, and hence the compounds do not interfere with cellular processes. Two representative examples of such dyes are given below:



37. What is the oxidation state of boron (B) in compound P?

38. The degree of unsaturation among the dyes (P) and (Q) which will preferentially stain fat tissues

Mathematics PART – III

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

39. P is any point, O being the origin. On the circle with OP as diameter the points Q and R are on the same side of the diameter such that $\angle POQ = \angle QOR = \theta$, if P, Q and R be complex numbers z_1, z_2, z_3 respectively such that $2\sqrt{3}z_2^2 = (2+\sqrt{3})z_1 z_3$ then the value θ is
(A) 30°
(B) 45°
(C) 15°
(D) 18°

40. For a function f: R → R, such that $f\left(\frac{2-x}{2}\right)+f\left(\frac{2+x}{2}\right) = 2f(1)$ for all $x \in R$ and $f(0) = 0$, then f(x) is
(A) even function
(B) odd function
(C) periodic
(D) none of these

41. If $\hat{a}, \hat{b}, \hat{c}$ are unit vectors, then the maximum value of $|\hat{a}+\hat{b}-\hat{c}|^2+|\hat{b}+\hat{c}-\hat{a}|^2+|\hat{c}+\hat{a}-\hat{b}|^2$ must be
(A) 9
(B) 12
(C) 15
(D) none of these

42. Let $A(\alpha, 0)$, $B(\beta, 0)$, $C(\gamma, 0)$, $D(\delta, 0)$ and α, β are the roots of equation $ax^2 + 2hx + b = 0$ while γ, δ are the those of $a_1x^2 + 2h_1x + b_1 = 0$ if C and D divides AB in the ratio $\lambda : 1$ and $\mu : 1$ respectively also ab_1, hh_1, a_1b are in A.P., then $\lambda + \mu$ is equal to
(A) 2
(B) 1
(C) 0
(D) none of these

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

43. If x_1 and x_2 are positive numbers between 0 and 1, then which of the following is/are true?
(A) $\sin\left(\frac{x_1+x_2}{2}\right) \le \frac{\sin x_1 + \sin x_2}{2}$
(B) $\tan\left(\frac{x_1+x_2}{2}\right) \le \frac{\tan x_1 + \tan x_2}{2}$
(C) $\log\left(\frac{x_1+x_2}{2}\right) \le \frac{\log x_1 + \log x_2}{2}$
(D) $\left(\frac{x_1+x_2}{2}\right)^2 \le \frac{x_1^2 + x_2^2}{2}$

44. Let $S_n (n \geq 1)$ be a sequence of sets defined by

$S_1 = \{0\}$, $S_2 = \left\{\frac{3}{2}, \frac{5}{2}\right\}$, $S_3 = \left\{\frac{8}{3}, \frac{11}{3}, \frac{14}{5}\right\}$

$S_4 = \left\{\frac{15}{4}, \frac{19}{4}, \frac{23}{4}, \frac{27}{4}\right\}$, … then

(A) third element in S_{20} is $\frac{439}{20}$

(B) third element in S_{20} is $\frac{431}{20}$

(C) sum of elements in S_{20} is 589

(D) sum of elements in S_{20} is 609

45. The lines $(m - 2) x + (2m - 5) y = 0$, $(m - 1) x + (m^2 - 7) y - 5 = 0$ and $x + y - 1 = 0$ are

(A) concurrent for three values of m

(B) concurrent for one values of m

(C) concurrent for no values of m

(D) are parallel for m = 3

46. Let the function f: $[-1, 1] \to R$ is defined by $f(x) = \int_{-1}^{x} |\sin \pi t - \cos \pi t| dt$, then

(A) f(x) is one-one function

(B) periodic function

(C) monotonic function

(D) $f(x) \in \left[0, \frac{4\sqrt{2}}{\pi}\right]$

47. Let $f(x) = (\ln(a^2 - 3a - 3)) |\sin x| + \left[\frac{a^2}{9}\right] \cos \pi x \ \forall \ x \in R$ (where [.] denote greatest integer function) be such that $f(x + T) = f(x)$, then

(A) if $T \in Q$, then $a \in \{-1, 4\}$

(B) if $T \notin Q$, then $a \in (-\infty, -1] \cup \{4\}$

(C) if $T \in Q$, then $a \in \left(-3, \frac{3+\sqrt{21}}{2}\right)$

(D) if $T \notin Q$, then $a \in \left(-3, \frac{3-\sqrt{21}}{2}\right)$

48. Let a, b, c, d > 0 and if $x = \frac{a}{a+b+d} + \frac{b}{a+b+c} + \frac{c}{b+c+d} + \frac{d}{a+c+d}$, then for some a, b, c, d, the value of

(A) $x = \frac{1}{2}$

(B) $x = \frac{5}{3}$

(C) $x = \frac{3}{2}$

(D) $x = \frac{5}{4}$

Section – B (Maximum Marks: 12)

*This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER.***

49. The polynomial $p(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17}$ can be written as a polynomial in y where $y = x + 1$, then let coefficient of y^2 be k then $\left[\frac{k}{400}\right]$ (where [.] denote greatest integer function) is_______

50. Let the area enclosed by the curve $|x - 60| + |y| = |x/4|$ be $4(n!)$ then the value of n is__________

51. Let f(n) denote the square of the sum of the digits of natural number n, where $f^2(n)$ denote f(f(n)), $f^3(n)$ denote f(f(f(n))) and so on. The value of $\frac{f^{2011}(2011) - f^{2010}(2011)}{f^{2013}(2011) - f^{2012}(2011)}$ = ___________

Section – C (Maximum Marks: 12)

*This section contains **THREE (03)** question stems. There are **TWO (02)** questions corresponding to each question stem. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Question Stem for Question Nos. 52 and 53

Question Stem

Let $f(x) = 2|x-3|, 1 \le x \le 5$ and for rest of the values $f(x)$ can be obtained by using the relation $f(5x) = \alpha f(x) \forall x \in R$.

52. The maximum value of $f(x)$ in $[5^4, 5^5]$ for $\alpha = 2$ is__

53. The value of $f(2007)$, taking $\alpha = 5$, is___

Question Stem for Question Nos. 54 and 55

Question Stem

Let $f(x)$ be a polynomial of degree 5 with leading coefficient unity, such that $f(1) = 5, f(2) = 4, f(3) = 3, f(4) = 2$ and $f(5) = 1$, then:

54. $f(6)$ is equal to____

55. Sum of the roots of $f(x)$ is equal to____

Question Stem for Question Nos. 56 and 57

Question Stem

Consider a triangle ABC with vertex $A(2,-4)$. The internal bisectors of the angles B and C are $x+y=2$ and $x-3y=6$ respectively. Let the two bisectors meet at I.

56. If (a,b) is incentre of the triangle ABC then $(a+b)$ has the value equal to___

57. If (x_1,y_1) and (x_2,y_2) are the co-ordinates of the point B and C respectively, then the value of $(x_1x_2+y_1y_2)$ is equal to____